Jesse Liu

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Appointments

Assistant Professor of Physics, New York University, USA	2025 – present
Junior Research Fellow, Trinity College, University of Cambridge, UK	2021 – 2024
Grainger Postdoctoral Fellow, University of Chicago, USA	2019 – 2021

Education

DPhil Particle Physics, University of Oxford, UK Thesis: 'Discovery strategies for dark matter and Higgsinos at the LHC' Supervisors: Alan Barr and John March-Russell	2015 – 2019
MSc Physics, Perimeter Institute and University of Waterloo, Canada Supervisor: Natalia Toro	2014 – 2015
BA Physics, University of Oxford, UK	2011 – 2014

Awards & Scholarships

Idea Day Prize, ATLAS Collaboration	2021
Junior Research Fellowship (4 years, deferred to Oct 2021), Trinity College, Cambridge	2019
Grainger Postdoctoral Fellowship (2 years, any field of physics), University of Chicago	2019
Best Poster (1st out of 226 entries), International Conference on High Energy Physics	2018
Travel Grant, International Conference on High Energy Physics	2018
Santander Academic Travel Award, University of Oxford	2018
STFC Postgraduate Studentship (3.5 years PhD scholarship)	2015
PSI Scholarship, Perimeter Institute and University of Waterloo	2014
Fitzgerald Prize, Exeter College, Oxford	2014
Summer Research Studentship, Oxford Particle Physics	2013
Gibbs Prize, winner of Oxford physics speaking competition	2013
East Scholarship, Exeter College, Oxford	2012
Fitzgerald Prize, Exeter College, Oxford	2012
Essay Prize, winner of Institute of Physics essay competition	2012

Papers

All papers available at INSPIREHEP. ATLAS author since 2017. Particle physics convention lists authors alphabetically. Selected papers with direct personal contributions:

- ATLAS ITk Strips Collaboration (**J. Liu** Section Editor), 'Mitigating Sensor Cracking in the ATLAS ITk Strip Detector by adding an Interposing Layer',
  JINST **21** (2026) P09052, arXiv:2609.10891

- ATLAS Collaboration (**J. Liu** Editor & Lead Analyzer), 'Measurement of Charged-Particle Production in $\sqrt{s_{\mathrm{NN}}} = 9.62$ TeV Proton-Oxygen Collisions as a Probe of Cosmic-Ray Air Showers with the ATLAS Detector', Phys. Rev. Lett. **137** (2026) 121901, arXiv:2604.05512
  ▷ *CERN News*, *Physics Briefing*, *EPP News*, *NYU News*, *The Conversation*, *Physics World*

- N. Fomin, B. Hommels, T. Ivison, K. Kariyapperuma and **J. Liu**, 'Extended thermal cycling of ATLAS ITk strip modules with and without stress mitigating interposers',
  JINST **20** (2025) P09036, arXiv:2507.12586
  ▷ *NYU EPP News feature*

- L. Beresford, S. Clawson and **J. Liu**, 'Strategy to measure tau $g-2$ via photon fusion in LHC proton collisions', Phys. Rev. D **110** (2024) 092016, arXiv:2403.06336

- S. Knirck, G. Hoshino, et. al. (BREAD Collaboration), 'First Results from a Broadband Search for Dark Photon Dark Matter in the 44 to 52 μeV Range with a Coaxial Dish Antenna', Phys. Rev. Lett. **132** (2024) 131004, arXiv:2310.13891
  ▷ *PRL Editors' Suggestion, UChicago News feature*

- ATLAS Collaboration (**J. Liu** Editor), 'Observation of the $\gamma\gamma \to \tau\tau$ Process in Pb+Pb Collisions and Constraints on the τ-lepton Anomalous Magnetic Moment with the ATLAS Detector', Phys. Rev. Lett. **131** (2023) 151802, arXiv:2204.13478
  ▷ *PRL Editors' Suggestion, ATLAS Physics Briefing, The Conversation feature*

- ATLAS Collaboration (**J. Liu** Chair of Editorial Board), 'Exclusive dielectron production in ultraperipheral Pb+Pb collisions at $\sqrt{s_{\rm NN}} = 5.02$ TeV with ATLAS', JHEP **06** (2023) 182, arXiv:2207.12781

- ATLAS Collaboration (**J. Liu** Member of Editorial Board), 'Measurement of the total cross section and ρ-parameter from elastic scattering in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector', Eur. Phys. J. C **83** (2023) 441, arXiv:2207.12246
  ▷ *On the Cover*, *ATLAS Physics Briefing*

- **J. Liu**, K. Dona, G. Hoshino, S. Knirck, N. Kurinsky, M. Malaker, D. W. Miller, A. Sonnenschein, M. H. Awida, P. S. Barry, K. K. Berggren, D. Bowring, G. Carosi, C. Chang, A. Chou, R. Khatiwada, S. Lewis, J. Li, S. W. Nam, O. Noroozian, T. X. Zhou, 'Broadband Solenoidal Haloscope for Terahertz Axion Detection', Phys. Rev. Lett. **128** (2022) 131801, arXiv:2111.12103
  ▷ *On the Cover and PRL Editors' Suggestion*, *UChicago News feature*

- K. Dona, **J. Liu**, N. Kurinsky, D. Miller, P. Barry, C. Chang, A. Sonnenschein, 'Design and performance of a multi-terahertz Fourier transform spectrometer for axion dark matter experiments', JINST **17** (2022) P06014, arXiv:2104.07157

- ATLAS Collaboration (**J. Liu** Editor & Lead Analyzer), 'Observation and Measurement of Forward Proton Scattering in Association with Lepton Pairs Produced via the Photon Fusion Mechanism at ATLAS', Phys. Rev. Lett. **125** (2020) 261801, arXiv:2009.14537
  ▷ *ATLAS Physics Briefing*, *CERN Courier feature*

- J. Amacker, W. Balunas, L. Beresford, D. Bortoletto, J. Frost, C. Issever, **J. Liu**, J. McKee, A. Micheli, S. Paredes Saenz, M. Spannowsky, B. Stanislaus, 'Higgs self-coupling measurements using deep learning in the $b\bar{b}b\bar{b}$ final state', JHEP **12** (2020) 115, arXiv:2004.04240

- ATLAS Collaboration, 'Searches for electroweak production of supersymmetric particles with compressed mass spectra in $\sqrt{s}$ = 13 TeV pp collisions with the ATLAS detector', Phys. Rev. D **101** (2020) 052005, arXiv:1911.12606

- L. Beresford and **J. Liu**, 'New physics and tau $g-2$ using LHC heavy ion collisions', Phys. Rev. D **102** (2020) 113008, arXiv:1908.05180
  ▷ *ParticleBites feature*

- I. Dawson, B. Gallop, **J. Liu**, P. Miyagawa, P. Phillips, G. Pownall, D. Robinson and A. Weidberg, 'In situ radiation damage studies of optoelectronics in the ATLAS SemiConductor Tracker', JINST **14** (2019) P07014

- L. Beresford and **J. Liu**, 'Search Strategy for Sleptons and Dark Matter Using the LHC as a Photon Collider', Phys. Rev. Lett. **123** (2019) 141801, arXiv:1811.06465

- ATLAS Collaboration (**J. Liu** Editor & Lead Analyzer), 'Search for electroweak production of supersymmetric states in scenarios with compressed mass spectra at $\sqrt{s} = 13$ TeV with the AT-

LAS detector', Phys. Rev. D **97** (2018) 052010, arXiv:1712.08119
▷ *Two ATLAS Physics Briefings*, *CERN Courier feature*, *Oxford Physics highlight*

- A. Barr and **J. Liu**, 'Analysing parameter space correlations of recent 13 TeV gluino and squark searches in the pMSSM', Eur. Phys. J. C **77** (2017) 202, arXiv:1608.05379
- A. Barr and **J. Liu**, 'First interpretation of 13 TeV supersymmetry searches in the pMSSM', arXiv:1605.09502, Published with 1608.05379 in Eur. Phys. J. C **77** (2017) 202
▷ *Oxford Physics highlight*

PROCEEDINGS

- **J. Liu** on behalf of the ALICE, ATLAS, CMS, LHCb Collaborations, 'Quantum Chromodynamics at the Large Hadron Collider', Proceedings of the Lepton Photon Symposium 2025, arXiv:2601.01217, ATL-PHYS-PROC-2025-114

INVITED SEMINARS

Cornell University, LEPP Journal Club Seminar, USA	9 Oct 2026
Brookhaven National Laboratory, Particle Physics Seminar, USA	28 May 2026
University of Chicago, Enrico Fermi Institute Colloquium, USA	11 May 2026
Rutgers University, High Energy Experimental Seminar, USA	23 Sep 2025
New York City College of Technology, Theoretical Physics Seminar, USA	27 Mar 2025
New York University, CCPP Brown Bag, USA	10 Mar 2025
University of Cambridge, Cavendish HEP Seminar, UK	7 May 2024
New York University, Experimental Particle Physics Seminar, USA	5 Mar 2024
University of California, Irvine, Particle Physics Seminar, USA	20 Feb 2024
Sorbonne Université, LPTHE Phenomenology Seminar, France	3 Oct 2023
Institut d'Astrophysique de Paris, ICAP Seminar, France (Online)	22 Jun 2023
University of Sydney, CPPC Seminar, Australia (Online)	4 May 2023
Johns Hopkins University, Experimental Particle Physics Seminar, USA	5 Apr 2023
Massachusetts Institute of Technology, LNS Colloquium, USA	27 Feb 2023
Brown University, High Energy Experiment Seminar, USA	15 Feb 2023
University of Edinburgh, Experimental Particle Physics Seminar, UK	27 Jan 2023
University of Bristol, Particle Physics Seminar, UK	7 Dec 2022
University of Liverpool, HEP Seminar, UK	16 Nov 2022
Imperial College London, HEP Seminar, UK	15 Jun 2022
University of Warwick, Particle Physics Seminar, UK	9 Jun 2022
Cornell University, LEPP Journal Club Seminar, USA (Online)	6 May 2022
Perimeter Institute for Theoretical Physics, Particle Physics Seminar, Canada	3 May 2022
Case Western Reserve University, Particle/Astro Seminar, USA (Online)	1 Mar 2022
University of Cambridge, Cavendish HEP Seminar, UK	30 Nov 2021
University of Tennessee, Knoxville, HEP Seminar, USA (Online)	8 Sep 2021
Cornell University, LEPP Journal Club Seminar, USA (Online)	30 Oct 2020
Stony Brook University, Particle Physics Seminar, USA (Online)	19 Oct 2020
University of Pittsburgh, PITT PACC Seminar, USA (Online)	5 Aug 2020
SLAC Stanford University, Particle Physics Theory Seminar, USA	12 Feb 2020
Fermilab, Cosmic Physics Center Chalk Talk, USA	14 Nov 2019
Fermilab, LPC Physics Forum, USA	31 Oct 2019
University of Cambridge, Cavendish HEP Seminar, UK	19 Feb 2019
LBNL University of California, Berkeley, RPM Seminar, USA	20 Nov 2018
SLAC Stanford University, Joint Theory–Experiment Seminar, USA	20 Apr 2018
Perimeter Institute for Theoretical Physics, BSM Seminar, Canada	17 Apr 2018
University of California, Santa Cruz, SCIPP Seminar, USA	10 Apr 2018

	University of Oxford, Dalitz Seminar, UK	19 Jan 2017
	University of Cambridge, Joint DAMTP–Cavendish Seminar, UK	13 Jan 2017
CONFERENCE TALKS	APS Division of Particles & Fields Meeting, Fermilab, USA	22 Jul 2026
	APS Global Physics Summit, Denver Convention Center, USA	16 Mar 2026
	Brookhaven Forum, Brookhaven National Laboratory, USA	23 Oct 2025
	CPAD Instrumentation Conference, University of Pennsylvania, USA	8 Oct 2025
	QCD@LHC Conference, Stony Brook University, USA	9 Sep 2025
	Lepton Photon Symposium (Invited Plenary), Madison, Wisconsin, USA	26 Aug 2025
	European Physical Society (EPS) HEP Conference, Marseille, France	7 Jul 2025
	Phenomenology Symposium, University of Pittsburgh, USA	19 May 2025
	Cavendish HEP Research Extravaganza, Cambridge, UK	6 Dec 2023
	CADEx Collaboration Meeting, Spain (Online)	16 Nov 2023
	BREAD Collaboration Meeting, University of Chicago, USA	5 Oct 2023
	EPS-HEP Conference, Hamburg, Germany	24 Aug 2023
	Dark Matter Beyond the Weak Scale Conference, Liverpool, UK	28 Mar 2023
	Photon-induced Processes Workshop, Durham, UK	3 Nov 2022
	Dark Matter UK Meeting, Imperial College London, UK	22 Sep 2022
	International Conference on High Energy Physics (ICHEP), Bologna, Italy	8 Jul 2022
	Institute of Physics (IoP) Joint APP HEPP Conference, Harwell, UK	5 Apr 2022
	Snowmass Cosmic Frontier CF2 Wavelike DM Meeting (Online)	8 Oct 2021
	APS Division of Particles & Fields Meeting, Florida State, USA (Online)	14 Jul 2021
	Phenomenology Symposium, Pittsburgh, USA (Online)	25 May 2021
	APS April Meeting (Online)	20 Apr 2021
	DIS Conference, Stony Brook, USA (Online)	14 Apr 2021
	LHC Forward Physics Meeting (Online)	4 Mar 2021
	Forward Physics and QCD with LHC, EIC and Cosmic Rays (Online)	23 Jan 2021
	Snowmass Energy Frontier EF09 BSM Meeting (Online)	4 Sep 2020
	ICHEP Conference, Prague, Czech Republic (Online, 2 talks)	30 Jul 2020
	Snowmass Energy Frontier Joint Preparatory Session (Online)	8 Jul 2020
	Higgs Couplings Conference, Oxford, UK	2 Oct 2019
	Young Experimentalists & Theorists Institute, Durham, UK	8 Jan 2019
	LHC Forward Physics Workshop, CERN, Switzerland	18 Dec 2018
	Oxford–Durham Di-Higgs Workshop, Durham, UK	6 Nov 2018
	SUSY Conference, Barcelona, Spain	23 Jul 2018
	ICHEP Conference (Plenary prize talk), Seoul, South Korea	6 Jul 2018
	DM@LHC Workshop, Heidelberg, Germany	5 Apr 2018
	Institute of Physics Joint APP HEPP Conference, Bristol, UK	27 Mar 2018
	Young Theorists Forum, Durham, UK	10 Jan 2018
	Young Theorists Forum, Durham, UK	11 Jan 2017
	2nd LHC BSM Reinterpretation Workshop, CERN, Switzerland	14 Dec 2016
	Theoretical Particle Physics Summer School, Manchester, UK	31 Aug 2016
	1st LHC BSM Reinterpretation Workshop, CERN, Switzerland	17 Jun 2016
COLLABORATION PLENARIES	ATLAS SM Group Plenary, CERN, Switzerland	5 Mar 2026
	ATLAS SM Soft QCD Plenary, CERN, Switzerland	1 Dec 2025
	ATLAS US Summer Workshop, Ann Arbor, Michigan, USA	23 Jul 2025
	ATLAS SM Electroweak Plenary, CERN, Switzerland	4 Oct 2024
	ATLAS UK SM+BPHYS Meeting, Oxford, UK	2 Oct 2024

ATLAS SM Workshop, Ljubljana, Slovenia	3 Sep 2024
ATLAS Week Plenary on ITk Strips, Thessaloniki, Greece	20 Jun 2024
ATLAS HV Breakdown Task Force Meeting (Online)	16 Feb 2024
ATLAS SM Workshop, Prague, Czechia	12 Sep 2023
ATLAS HV Breakdown Task Force Meeting (Online)	28 Jul 2023
ATLAS ITk Strips Module Meeting (Online)	18 Jul 2023
ATLAS UK Meeting, Manchester, UK	9 Jan 2023
ATLAS SUSY Workshop, CERN, Switzerland	19 Sep 2022
ATLAS ITk WP12 Meeting, Birmingham, UK	31 Mar 2022
ATLAS Upgrade Physics (Online)	22 Apr 2021
ATLAS SM Plenary (Online)	15 Apr 2021
ATLAS Run 4 Roman Pot Meeting (Online)	26 Mar 2021
ATLAS Idea Day (Online)	20 Jan 2021
ATLAS Roman Pot General Meeting (Online)	7 Oct 2020
ATLAS SM Jamboree (Online)	30 Sep 2020
ATLAS SM Plenary Analysis Closure (Online)	2 Jul 2020
ATLAS Week Plenary on SM & AFP (Online)	24 Jun 2020
ATLAS Roman Pot General Meeting, CERN, Switzerland	23 Jan 2020
ATLAS Hadronic Final State Forum, Chicago, USA	20 Dec 2019
ATLAS UK SUSY–Exotics Meeting, Cambridge, UK	11 Apr 2019
ATLAS SUSY Run 3 Ideas, CERN, Switzerland	22 Nov 2018
ATLAS SUSY Plenary, CERN, Switzerland	19 Jul 2018
ATLAS Exotics Workshop, Rome, Italy (poster)	29 May 2018
ATLAS Week Plenary on SUSY, CERN, Switzerland	21 Feb 2018
ATLAS Analysis Open Presentation, CERN, Switzerland	1 Nov 2017
ATLAS Exotics–SUSY Workshop, Bucharest, Romania	12 May 2017
ATLAS SUSY Electroweak, CERN, Switzerland	10 Feb 2017
ATLAS UK Meeting, Liverpool, UK	5 Jan 2017
ATLAS Week Plenary on SCT/ID Gen, CERN, Switzerland	17 Oct 2016

MENTORING & TEACHING

Postdoctoral researcher mentoring

Sebastian Rutherford Colmenares, NYU	2025 – present

Graduate student mentoring

Chris Sellgren, NYU	2026 – present
Christina Dorofeev, NYU	2026 – present
Amelia Stevens, NYU	2025 – present
Anna Mullin, Cambridge	2022
Kristin Dona, UChicago	2019 – 2021
Savannah Clawson, Manchester	2019 – 2020
Santiago Paredes Saenz, Oxford	2018 – 2020
Beojan Stanislaus, Oxford	2018 – 2020
Gabriel Gallardo, Oxford	2018 – 2019
Gavin Pownall, Oxford	2018 – 2019

Undergraduate student mentoring

Liliana Valle, REU summer student, UChicago	2021
Joe Ray, MPhys student, Oxford	2018 – 2019
Dan Kreso, MPhys student, Oxford	2017 – 2018
Alessandro Micheli, summer student, Oxford	2018

Ewan McCullough, summer student, Oxford 2017
Charles Sillett, summer student, Oxford/CERN 2017

New York University courses
PHYS-UA 123 Undergraduate Quantum Mechanics I Fall 2026
PHYS-GA 2027 Graduate Particle Physics Spring 2025, 2026

Member of PhD Thesis Defense Committees, NYU Physics
Zubair Bhatti, Matthew Drnevich 2026

Member of PhD Oral Qualification Exam (1 student), NYU Physics 2026
Panelist, 'Working with your PhD supervisor' workshop, Trinity College 2023
Trinity College Postdoc-PhD mentoring programme 2022
Lecturer, Oxford undergraduate summer students: 'The Standard Model' Jul 2018
Tutor, Oxford 3rd year undergraduates: 'Flows, Fluctuations & Complexity' May 2016
Tutor, Oxford 3rd year undergraduates: 'Subatomic Physics' May 2016

Leadership & Service

Inner Tracker Strip Module Level 3 Manager (Co-PI), US ATLAS Upgrade 2026 – present
▷ Project management of 27M$ Department of Energy detector construction
▷ Control Account Manager for production of 6.4k modules at three US assembly sites

Convener of QCD physics sessions, LHCP Conference, Paris, France 2025 – 2026
Convener of QCD physics sessions, EPS-HEP Conference, Marseille, France 2025
▷ Invited co-organizer of agenda, abstract reviewer, and chair for parallel talk sessions

Convener for LHC Physics Centre at CERN (LPCC) Forward Physics Group 2023 – 2025
▷ ATLAS contact, co-organizer, and convener for March, July and December 2024 workshops

Cambridge ATLAS Inner Tracker (ITk) strips upgrade module production 2021 – 2024
▷ Qualified cleanroom site for module production, mechanical and electrical quality control
▷ Led assembly and quality control of 60+ ITk strips modules to prove production readiness
▷ Module thermal cycling: pioneered electromechanical stress analysis and interposer solution

Convener of ATLAS Standard Model Soft QCD physics group 2022 – 2024
▷ Coordination role organizing program of 20 analyses probing non-perturbative dynamics
▷ Chaired bi-weekly plenary meetings, steered physics group strategy and workshop agendas

Convener for Run 4 AFP Task Force on HL-LHC physics projections 2021 – 2022
▷ Coordination role organizing physics and performance for ATLAS Forward Proton upgrade
▷ Chaired meetings discussing beam optics, detector performance and physics opportunities

ATLAS Forward Proton combined performance and data preparation 2019 – 2020
▷ Led scrutiny of AFP data quality, detector efficiency, energy scale calibration, simulation
▷ Delivered first collaboration-wide AFP performance recommendations for 2017 dataset

Colloquium committee member, NYU Physics 2026 – present
Institute Representative (Team Leader), NYU ATLAS 2025 – present
Editorial Board member for ATLAS SM Soft QCD analysis STDM-2021-01 2025 – present
Analysis Contact, ATLAS Standard Model photon fusion production in Run 3 2024 – present
Institute paper review of EXOT-2021-33, NYU ATLAS May 2026
Computer and Departmental Webpage committee, NYU Physics 2025 – 2026
Contact Editor, ATLAS SM proton-oxygen analysis STDM-2025-08 2025 – 2026
Graduate physics open house talk, NYU Department of Physics 2026, 2025
Institute paper review of STDM-2018-41, NYU ATLAS Jan 2025
Invited member-at-large for subgroup convenors, ATLAS Standard Model 2024

	Grant expert reviewer for National Science Centre, Poland	2023
	Journal referee for Annalen der Physik	2023
	ATLAS paper review (Institute Reading), Heavy-ion and Higgs, Cambridge ATLAS	2023
	Contributing author, Snowmass whitepaper 'Axion Dark Matter', 2203.14923	2022
	UChicago ATLAS website redesign, authored recruitment and mission statements	2021
	Journal referee for Nuclear Physics B	2021
	ATLAS paper review (Institute Reading), Trigger, UChicago ATLAS	Dec 2019, Sep 2020
	Coordinator for Snowmass Letter of Interest on terahertz axions	Aug 2020
	Grainger Fellowship website development, UChicago Physics	Jul 2020
	Postdoctoral community working group, UChicago KICP	Jul 2020
	Journal club, UChicago ATLAS group	May 2020
	Graduate open day speaker & poster, Oxford Particle Physics	Dec 2018
	ATLAS paper review (Institute Reading), Supersymmetry, Oxford ATLAS	Jul 2018
	Organising committee member, YETI 2017, Durham University	Sep 2016
	Physics aptitude test marker, Oxford undergraduate admissions	Nov 2015
	Resident advisor for US visiting students, Williams-Exeter Programme at Oxford	2012 – 2014
	Publicity officer, Oxford Undergraduate Physics Conference	Mar 2013

OUTREACH	Beyond Standard and ATLAS Blog, explainer via US-CERN communications	9 Oct 2026
	Washington Square News, interviewed by student paper on cosmic-ray physics	29 Sep 2026
	Physics World, interviewed by science journalist for research news article	21 Sep 2026
	The Conversation, pitched and authored news explainer on cosmic-ray result	14 Sep 2026
	NYU News Release, proposed and coordinated ATLAS proton-oxygen coverage	14 Sep 2026
	ATLAS Physics Briefing, co-authored cosmic-ray news with ATLAS Outreach	15 Apr 2026
	NYU EPP News, wrote news article on proton-oxygen cosmic-ray result	10 Apr 2026
	Faculty Science Communication Workshop, NYU Carter Journalism Institute	Oct–Nov 2025
	NYU Conference for Undergraduate Women in Physics, first-gen panelist	25 Jan 2025
	The Conversation, news article with science journalists on taus in heavy ions	17 Oct 2023
	Sidney Sussex MCR Research Conference, Cambridge, invited guest speaker	4 Mar 2023
	Interviewed for UChicago news article on BREAD axion experiment proposal	12 Apr 2022
	ATLAS Physics Briefing, news article on taus in heavy ions result	7 Apr 2022
	Trinity College Science Society Symposium, 'Dark matter and colossal colliders'	13 Mar 2022
	CERN Courier magazine article, 'The LHC as a photon collider'	15 Aug 2020
	ATLAS Physics Briefing, online news article on AFP result	30 Jul 2020
	Trinity Forum, Cambridge, 'Dark matter at the LHC' invited talk	5 Jun 2019
	Pint of Science, Manchester, 'Using light to reveal the darkness' invited talk	20 May 2019
	Oxford HiggsHunters, mentoring secondary school students	26–27 Jun 2018
	CERN school visit, hosting 35 sixth form students, teacher engagement	15 Feb 2018
	FQXi essay, information in physics and biology	3 Mar 2017
	Lunch with CERN scientists, physics discussions with school students	21 Oct 2016
	Oxfordshire Science Festival, all-day live physics demos on Broad Street	26 Jun 2016
	Perimeter Institute PSI Blog, 'Surprises of spin'	15 May 2016
	Out of this World: Family Fun Day, targeting under-represented at university	17 Apr 2016
	Oxford Catz Exchange, 'Seeing the unseeable' talk	25 Feb 2016
	Oxford FameLab, 'Listening to numbers' talk	25 Jan 2016
	Oxford Stargazing Live, all-day live demos at particle physics stall	16 Jan 2016
	Oxford Science Christmas Lectures, mentoring school students	8 Dec 2015
	Oxford Flash Talk Physics, 'Fiddling with physics' invited talk	4 Dec 2013
	Institute of Physics essay, 'Predicting the unpredictable'	30 May 2012